PAPER_560: Buoyancy-Stratified Factorial Geometry — Holonomy Group and Parallel Transport

Unified Quantum Field Framework (UQFF)

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PAPER_560: Buoyancy-Stratified Factorial Geometry — Holonomy Group and Parallel Transport

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> ***Context note:*** *The holonomy group of a manifold records how vectors rotate under parallel transport around closed loops. PAPER_554 showed BSFG has non-zero Riemann curvature $R^r{}_{0r0} \neq 0$; PAPER_556 showed the 22 extra dimensions compactify to flat tori T^{22}. This paper combines both to determine the complete holonomy group $G_{\rm hol}(\mathcal{M}^{26})$ and derives the parallel transport phase $\delta\phi$ for a given loop area.*

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§1 Abstract

We determine the holonomy group of the 26-dimensional BSFG manifold $\mathcal{M}^{26} = \mathcal{M}^4_{\rm BSFG} \times T^{22}$. The 4D BSFG slice has Ricci scalar $R_{\rm scalar} \neq 0$ — it is not Ricci-flat — so it carries the generic pseudo-Riemannian holonomy $SO^+(3,1)$. The 22 compactified extra dimensions (PAPER_556) form flat tori with holonomy $U(1)^{22}$. The full result:

$$\boxed{G_{\rm hol}(\mathcal{M}^{26}) = SO^+(3,1) \times U(1)^{22}}$$

Exceptional holonomies G_2 and $Spin(7)$ are rigorously excluded. The parallel transport angle around a small loop of coordinate area ΔA:

$$\delta\phi = R^r{}_{0r0} \cdot \Delta A = \frac{6\eta\cos(\pi t_n)C_{\rm num}}{r^5} \cdot \Delta A$$

At a Planck-area loop (l_P^2): $\delta\phi_P \approx 4.07 \times 10^{-89}$ rad — completely unobservable at current precision.

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§2 Holonomy Decomposition

PAPER_556 established the 26D line element:

$$ds^2_{26} = A_{\mu\nu}dx^\mu dx^\nu + \sum_{i=5}^{26} L_i^2(r)\,d\theta_i^2$$

with $L_i(r) \to 0$ for $r \gg r_P$ (full compactification at all observed scales). The manifold thus factorizes:

$$\mathcal{M}^{26} \simeq \mathcal{M}^4_{\rm BSFG} \times T^{22} \qquad (r \gg r_P)$$

Proposition: The holonomy of a product manifold $M_1 \times M_2$ is $G_{\rm hol}(M_1) \times G_{\rm hol}(M_2)$.

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§3 Holonomy of the 4D BSFG Slice

Berger's classification theorem (1955) states: for an irreducible, simply-connected, complete Riemannian manifold that is not a symmetric space, the holonomy group is one of:

$$SO(n),\; U(n/2),\; Sp(n/4),\; G_2\ (n=7),\; Spin(7)\ (n=8)$$

with the exceptional cases G_2 and $Spin(7)$ requiring **Ricci-flatness** ($R_{\mu\nu} = 0$).

Step 1. From CP4 #149: $R_{\rm scalar}(R_\odot) \approx 3.12 \times 10^{-19}\ {\rm m}^{-2} \neq 0$.

Step 2. Therefore $R_{\mu\nu} \neq 0$, so $\mathcal{M}^4_{\rm BSFG}$ is **not Ricci-flat**.

Step 3. For a 4D Lorentzian manifold (pseudo-Riemannian, signature $+{-}{-}{-}$) the corresponding holonomy is $SO^+(3,1)$ (the restricted Lorentz group) in the generic non-symmetric case.

Exceptional group	Required condition	BSFG status
G_2	$\dim = 7$, Ricci-flat	✗ wrong dim, $R \neq 0$
$Spin(7)$	$\dim = 8$, Ricci-flat	✗ wrong dim, $R \neq 0$

$$\boxed{G_{\rm hol}(\mathcal{M}^4_{\rm BSFG}) = SO^+(3,1)}$$

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§4 Holonomy of the Compactified Sector

The 22 extra dimensions (i = 5, …, 26) are compactified via $L_i(r) = r_P \exp(-r^i/(i!\,r_P^{i-1}))$. At $r \gg r_P$, each circle S^1_i has curvature $\kappa_i = 1/L_i \to \infty$ — effectively flat at macroscopic scales. A flat torus T^1 has holonomy $U(1)$.

$$G_{\rm hol}(T^{22}) = U(1)^{22}$$

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§5 Full BSFG Holonomy

$$\boxed{G_{\rm hol}(\mathcal{M}^{26}) = SO^+(3,1) \times U(1)^{22}}$$

Generators: 6 from $SO^+(3,1)$ (Lorentz boosts + rotations) + 22 from $U(1)^{22}$ = **28 generators total**.

Note: This is consistent with but distinct from the isometry group $G_{\rm iso} = SO(3) \times U(1)^{23}$ (26 generators, CP4 #152). The holonomy group relates to the curvature connection; the isometry group relates to Killing symmetries.

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§6 Parallel Transport Formula

The Ambrose–Singer theorem relates the holonomy algebra to the curvature 2-form. For an infinitesimal closed loop with coordinate area element ΔA in the (r, t) plane:

$$\delta\phi^r{}_0 = R^r{}_{0r0} \cdot \Delta A = \frac{6\eta\cos(\pi t_n)C_{\rm num}}{r^5} \cdot \Delta A$$

Step 6. Values at $r = R_\odot$, $t_n = 0$:

Loop area ΔA	$\delta\phi$ (rad)
$l_P^2 = (1.616 \times 10^{-35})^2\ {\rm m}^2$	$\approx 4.07 \times 10^{-89}$
$R_\odot^2 = (6.96 \times 10^8)^2\ {\rm m}^2$	$\approx 7.53 \times 10^{-2}$
$1\ {\rm AU}^2 = (1.496 \times 10^{11})^2\ {\rm m}^2$	$\approx 8.0 \times 10^{-4}$ (eval. at 1 AU)

The $R_\odot^2$ result (~ 0.075 rad $\approx 4.3°$) shows BSFG parallel transport is detectably non-trivial at solar scales — a loop spanning the solar disk accumulates ~4° of holonomy rotation.

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§7 References

- CP4 #149 — `BSFGRiemannCurvatureAetherMetricCalculator` — PAPER_554 (Riemann $R^r{}_{0r0}$)
- CP4 #151 — `BSFG26DLineElementFactorialCompactificationCalculator` — PAPER_556 (T^{22} compactification)

- CP4 #152 — `BSFGSymmetryGroupIsometryAnalysisCalculator` — PAPER_557 (26 Killing generators)
- Berger, M. (1955). *Sur les groupes d'holonomie homogène des variétés à connexion affine et des variétés riemanniennes.* Bull. Soc. Math. France.